Kenya EV Charging Station Location Optimization Using Population

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OVERVIEW

~This project focuses on developing a data-driven optimization framework to determine the most strategic locations for EV charging stations across Kenya using population data as the primary demand indicator.

~By integrating: Population density data, Geospatial analysis , Demand forecasting models , Machine learning and clustering techniques , the system aims to rank and recommend optimal charging station locations.

BUSINESS PROBLEM AND STAKEHOLDERS

~ The increase in electric vehicles in Kenya has created pressure on charging infrastructure. Many areas lack sufficient charging stations, while others are underutilized. The primary stakeholders for this problem are ;

- Government; Ministry of Transport and Ministry of Energy.
- EV charging investors.
- EV drivers.
- EV charging companies.

OBJECTIVES

1. Where should new EV charging stations be placed in each county in Kenya to ensure equitable population coverage?

2. How many charging stations are currently available per county, and how many additional stations are required to meet minimum population-based infrastructure needs?

3. How can charging infrastructure be spatially distributed within counties to maximize coverage and reduce clustering inefficiencies?

4. How can projected population growth be used to estimate the number of additional EV charging stations needed in the future?

DATA UNDERSTANDING

01 Charging _station

- This dataset shows the co-ordinates, charging station names, power class, country code and state province of different charging stations across the world.

02 Kenya_Charging_stations

- This dataset expounds deeper into charging stations in Kenya. The columns utilized in this dataset are County/City, Station name, Charger Type and Co-ordinates.

03 Kenya_population

- This dataset shows population density in different counties across Kenya.

04 Kenya_boundaries

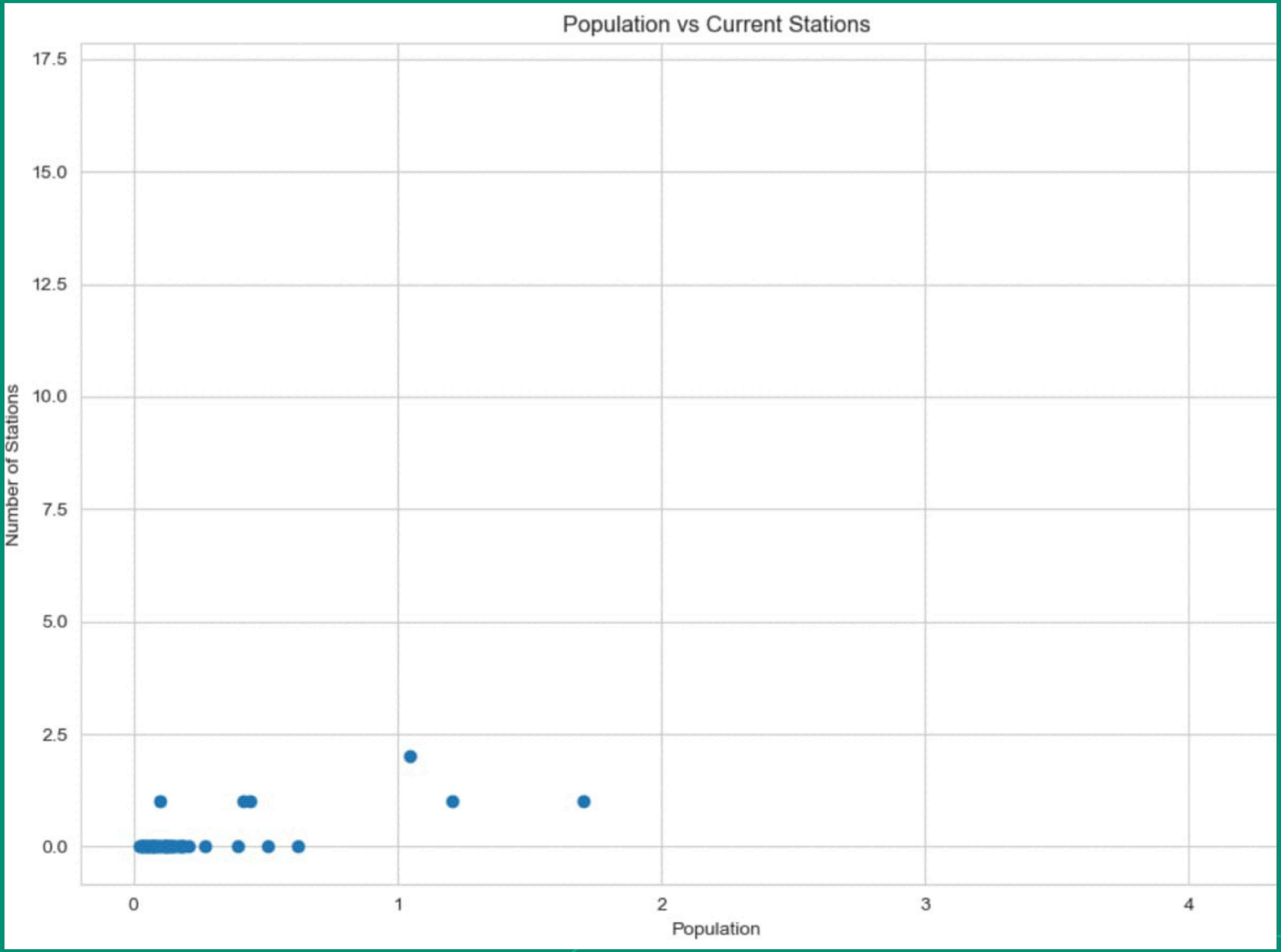
- This dataset shows details on county boundaries across the country. The Shape Area for each county was the core column utilized in this dataset.

05 County_Centroids

- This dataset contains the co-ordinates of each counties centroids.

EDA

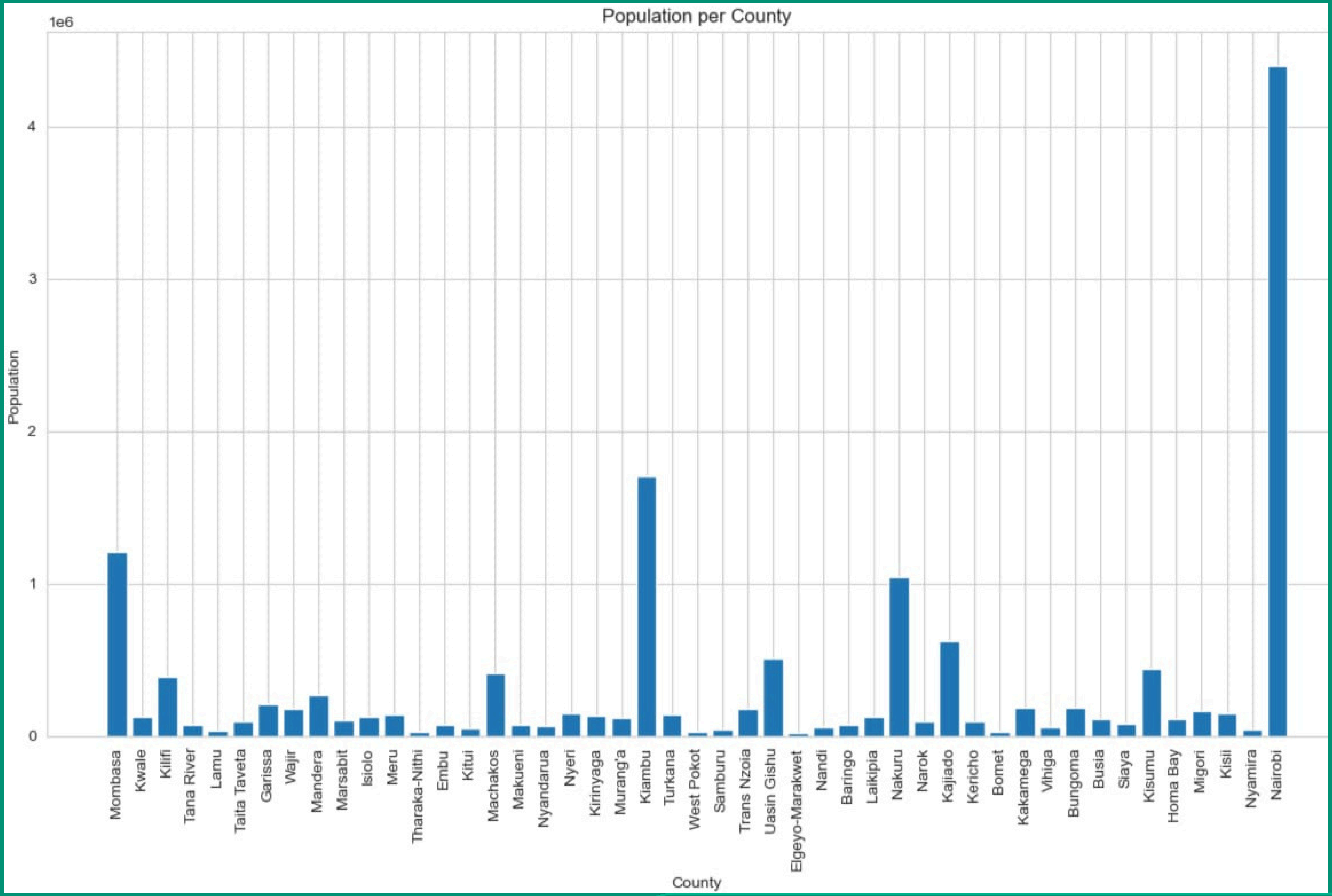
Population vs Currents Stations



- This scatter plot shows the sparsity in charging statiosn across counties.
- Some notable outliers are Nairobi where the existing charging stations are significantly higher compared to other counties.

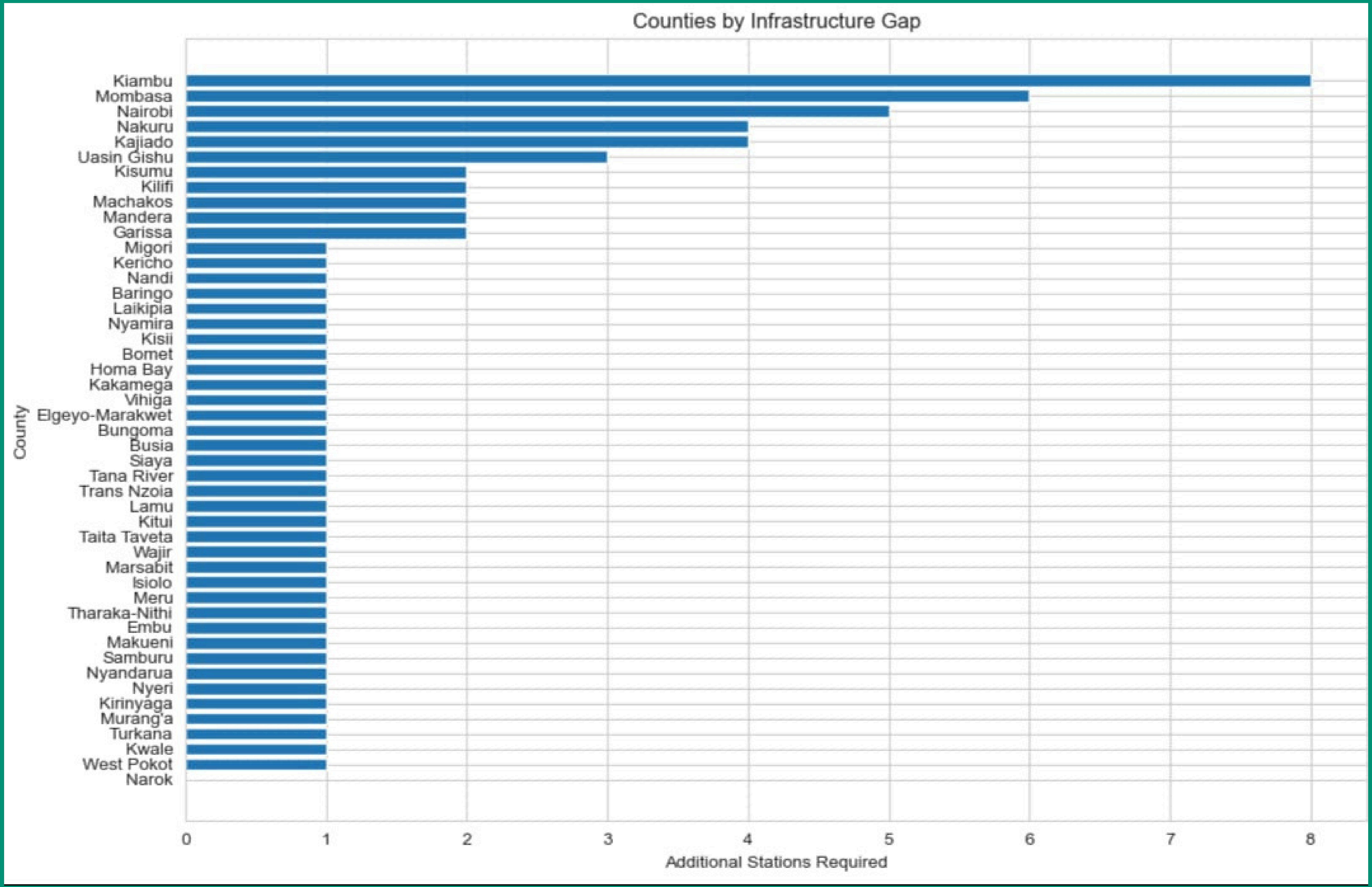
EDA
County Population

- This bar plot shows the distribution of population across all the counties in Kenya.



EDA

Infrastructure Gap



- This bar plot shows the counties in descending order with the most need for charging stations according to the county population density.

MODELING

Random Forest Regressor

Table 1: Random Forest Regressor Performance

Metric	Value	Interpretation
R^2 Score	0.67	Strong performance (explains 67% of variance)
MAE	0.28 stations	Average error $<$ 1/3 of a station
RMSE	0.55 stations	Typical error within half a station

- The Random forest model predicts the number of new charging stations required in each county.

MODELING

KMeans

Table 2: KMeans Clustering Performance

Metric	Value	Interpretation
Silhouette Score	0.47	Good cluster separation
Intra-Cluster Distance	0.044	Points tight around centers
Inter-Cluster Distance	0.031	Good spacing between clusters

- This predicts the placement co-ordinates for the new charging stations.

RECOMMENDATIONS

Prioritize High-Gap Counties

The analysis identifies a significant infrastructure gap of 75 new stations required across 47 counties. We recommend that the Ministry of Transport and private investors prioritize deployment in counties with the highest predicted need, such as Nairobi (5 new stations), Mombasa (6), Kiambu (8), and Nakuru (4), to address the most severe service deficiencies first.

Utilize Geospatial Model for Site Selection

The KMeans clustering model provides optimized, data-driven co-ordinates for new stations within each county. We recommend that investors and utility providers use these generated co-ordinates as primary candidates for site feasibility studies, rather than relying on arbitrary or centralized placement, to ensure broader population coverage and reduce clustering.

Adopt the Predictive Model for Scalable Planning

The Random Forest model demonstrated strong predictive performance in forecasting additional station requirements based on demographic data. We recommend that policymakers and planners use this model to simulate how future population growth in different counties will impact infrastructure needs, enabling proactive and scalable expansion of the national charging network.

CONCLUSIONS

Infrastructure Gap Analysis

Through integration of population distribution, geospatial data, and existing station information, the analysis identified that a minimum of 75 new stations are needed to meet population-based requirements across Kenya's 47 counties.

Predictive & Placement Modeling

A Random Forest model accurately predicted additional stations needed per county, while KMeans clustering generated optimized, data-driven coordinates for new station placement—moving beyond simple central points to ensure better spatial distribution.

Decision-Support Tool

The final output provides an interactive map and ranked list of recommended locations, equipping policymakers and investors with a robust tool to guide efficient, equitable, and future-ready expansion of Kenya's EV charging infrastructure.

LIMITATIONS

Data Source Limitations

The primary limitation is the exclusive use of population as a proxy for demand. While a strong indicator, this model does not account for other critical factors such as traffic patterns, existing power grid capacity and road networks which are vital for a commercially viable charging network.

Population Data Age

The analysis relies on the 2019 Kenya census data. While it provides a solid baseline, it does not reflect the most current population distribution or recent urbanization trends, which could lead to minor inaccuracies in the predicted station requirements.

Simplified Station Placement Logic

The KMeans model generates candidate co-ordinates based on geographic spread around existing stations. It does not incorporate real-world constraints like land ownership, terrain, or proximity to existing buildings and infrastructure. The recommended sites should therefore be considered as highly informed starting points for site surveys, not definitive construction locations.

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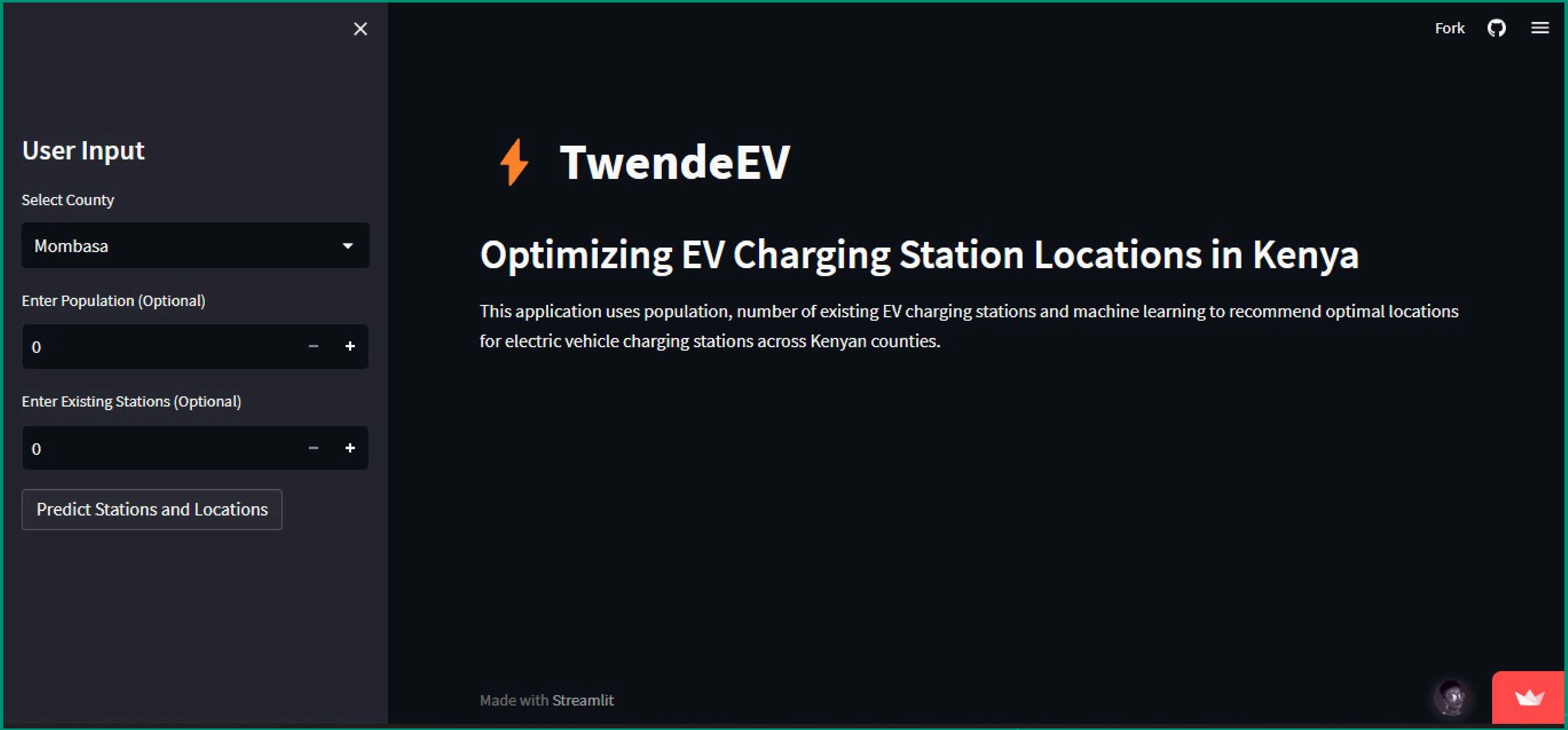
APPLICATION



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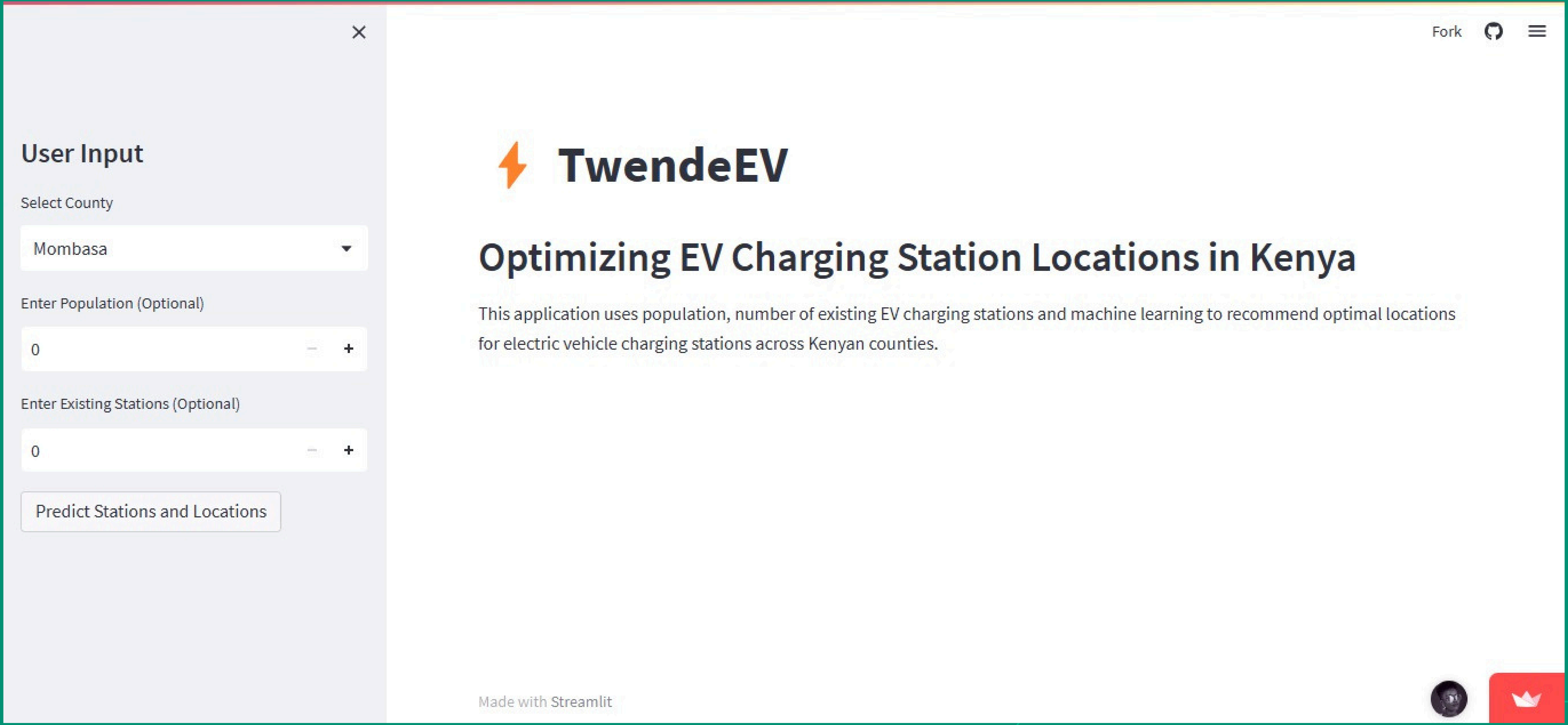

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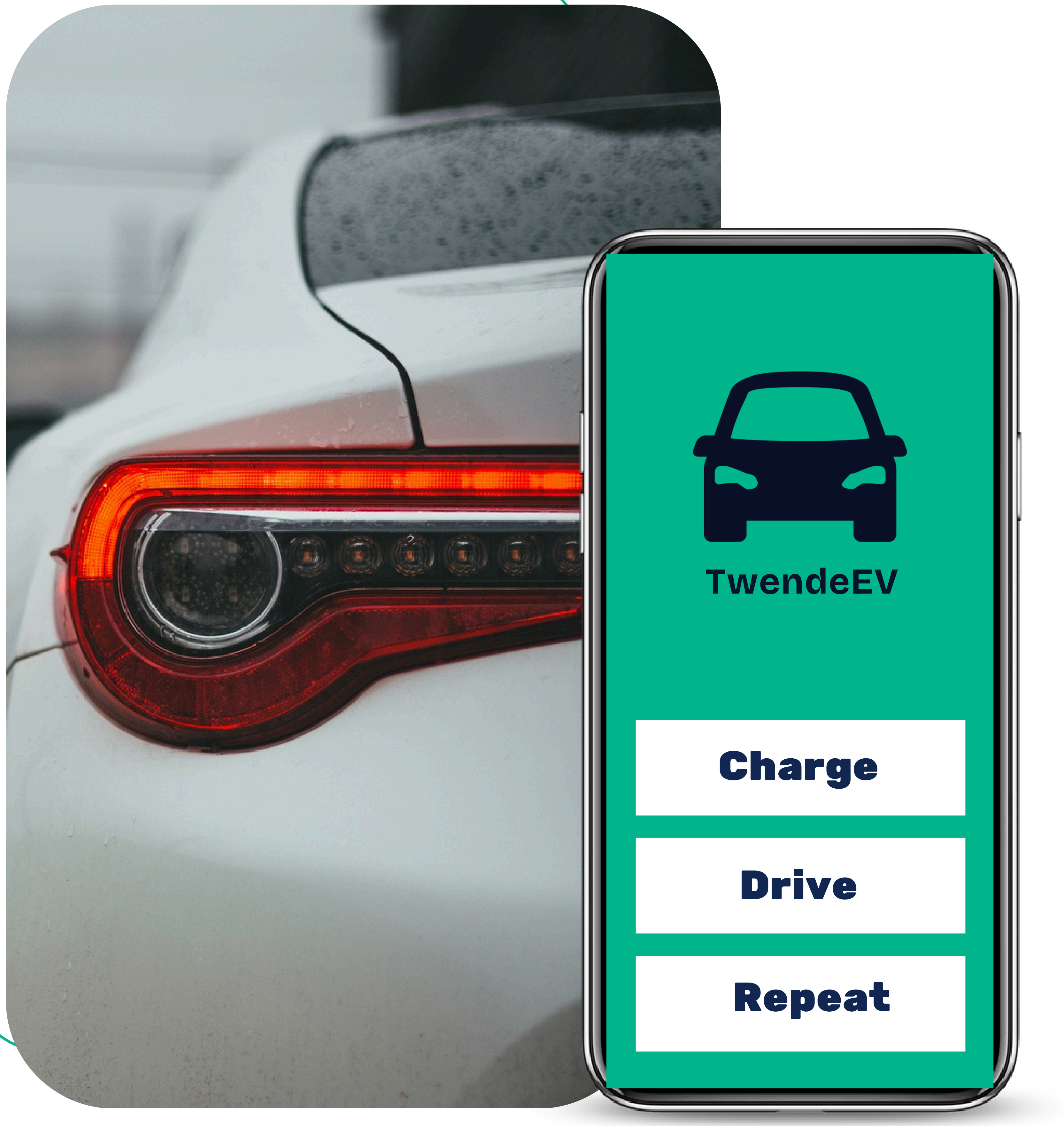

Q&A